Lecture 01:

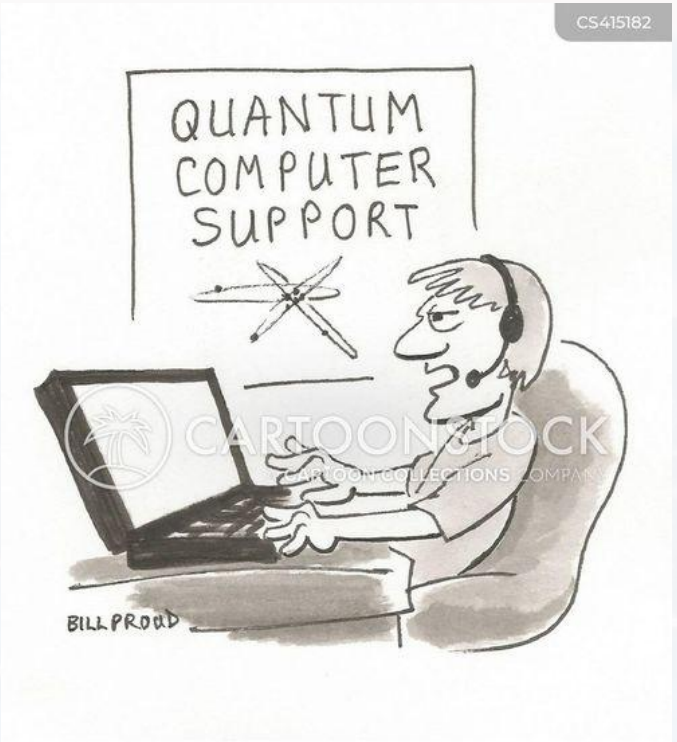
Introduction

Markus Hohle

University California, Berkeley

Numerical Methods for Computational Science

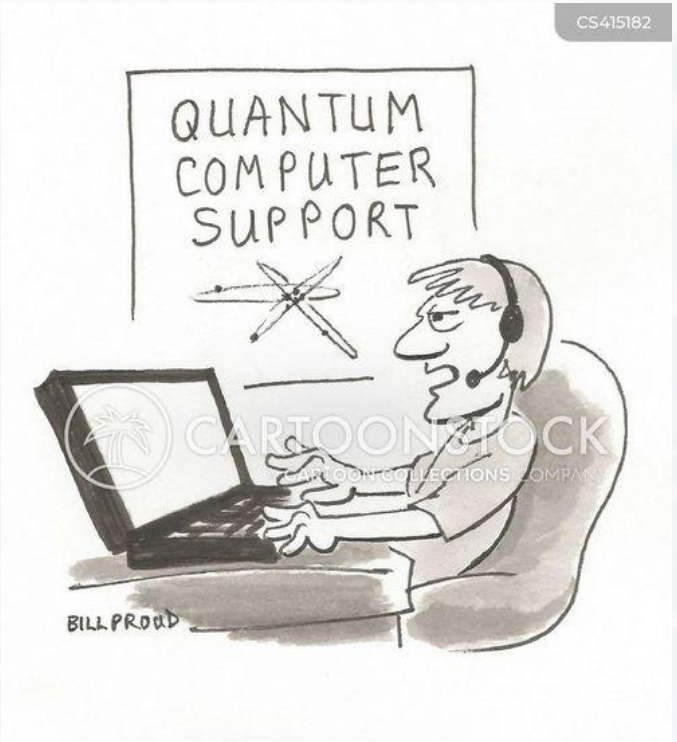
MSSE 273, 3 Units



"Try switching it off and on at the same time."

Outline

- Motivation
- Course Map
- Python Libraries
- Lecture Exercise



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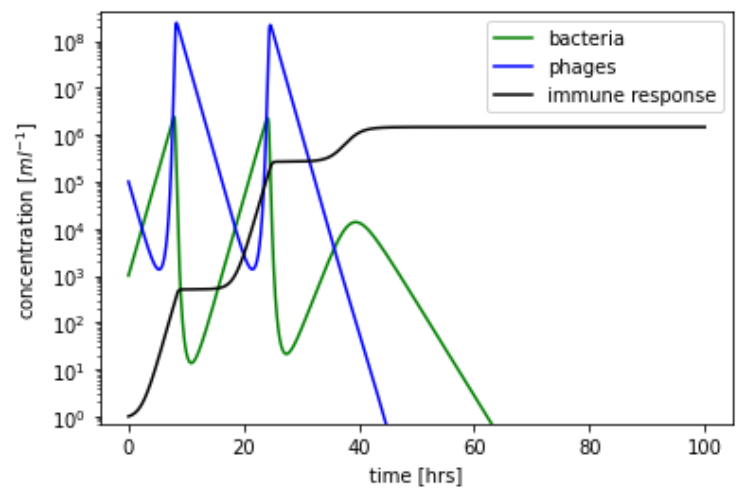
Why Numerical Methods?



Leung & Weitz 2016

drug development

AI driven data analysis (lecture exercise, see Chem 277B)

```
>BEISA025-19|Culex|COI-5P
AACATTATATTTCATTTTTGGTGCTTGAGCAGGAATAATTGGAACTTCTTTAAGTCTTCTTATTCG
AGCTGAATTAAGTCAACCAGGAGTTTTTATTGGGAATGATCAAATTTATAATGTAATTGTTACAGC
TCATGCTTTTATTATAATTTTTTTTATAGTAATACCTATTATAATTGGAGGATTTGGAAATTGATT
AGTTCCTTTAATACTAGGAGCTCCTGATATAGCTTTTCCTCGAATAAATAATATAAGATTTTGAAT
ACTTCCCCCCTCATTAACACTTCTACTTTCTAGTAGTATAGTAGAAAATGGAGCTGGTACAGGTTG
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TTCTTTACATTTAGCCGGAATTTCTTCAATTTTAGGAGCTGTAAATTTTATTACTACTGTAATTAA
TATGCGATCTTCTGGTATTACCCTTGATCGAATACCTTTATTTGTTTGATCAGTTGTAATTACTGC
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```
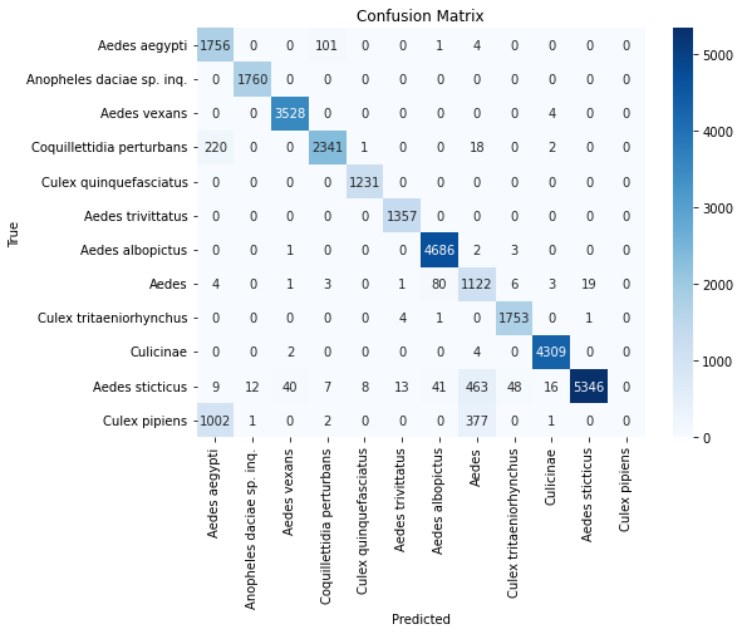
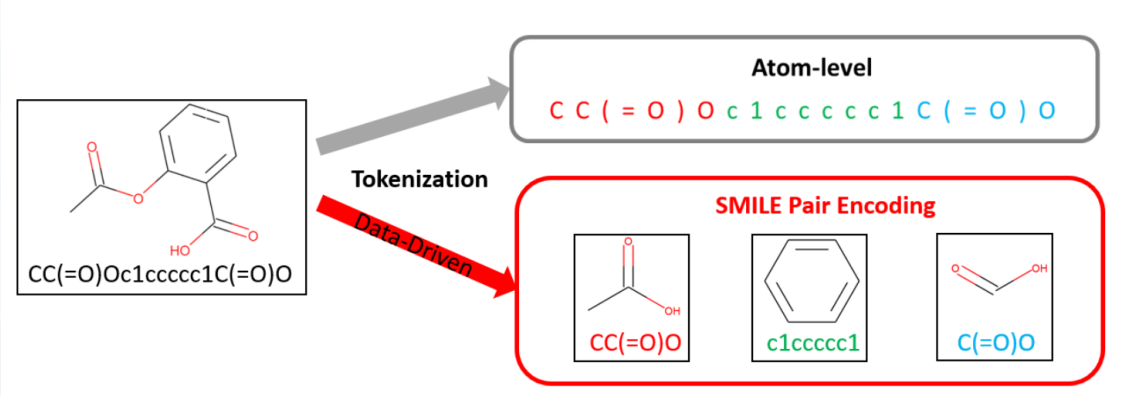
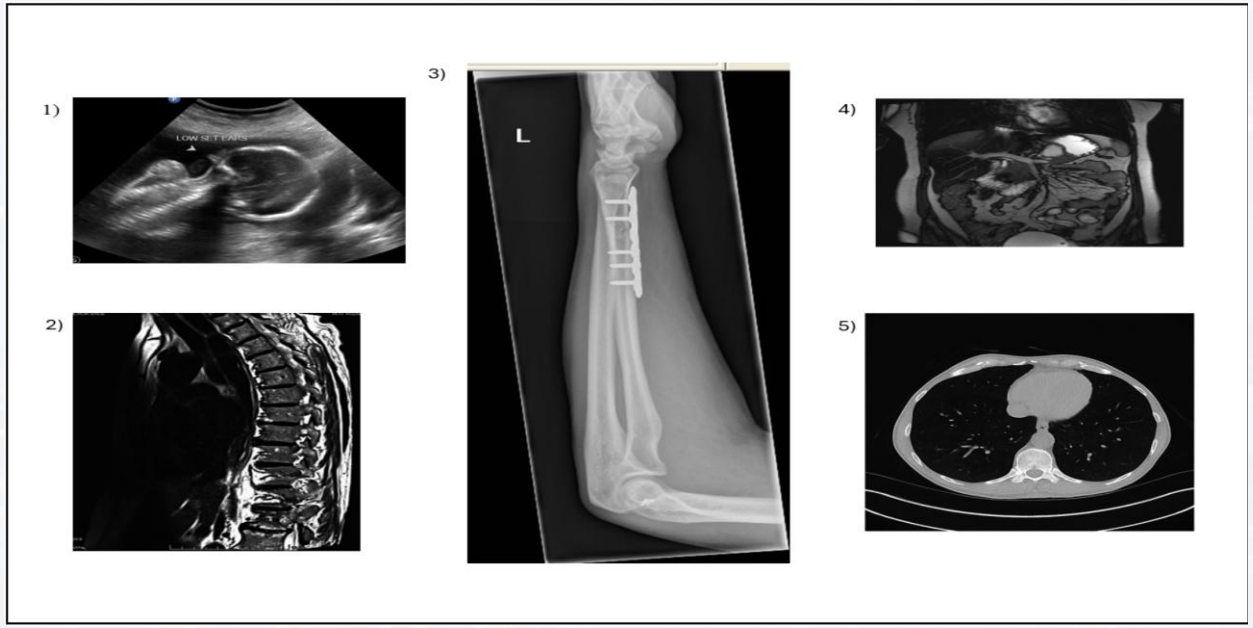



Why Numerical Methods?

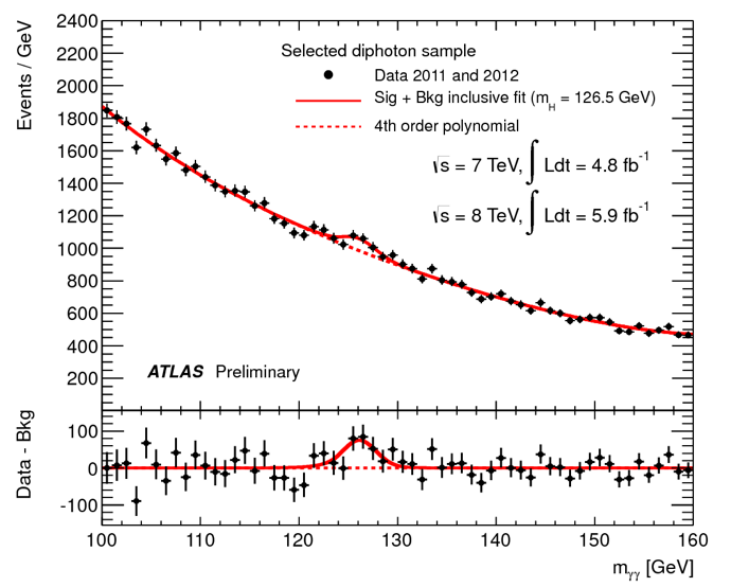


automating workflows (Mass Spec to SMILES, Chem 277B project, **Casey Tomlin, Dulce Torres, Esther Mathew, Jesse Maki, Marie Anand**)

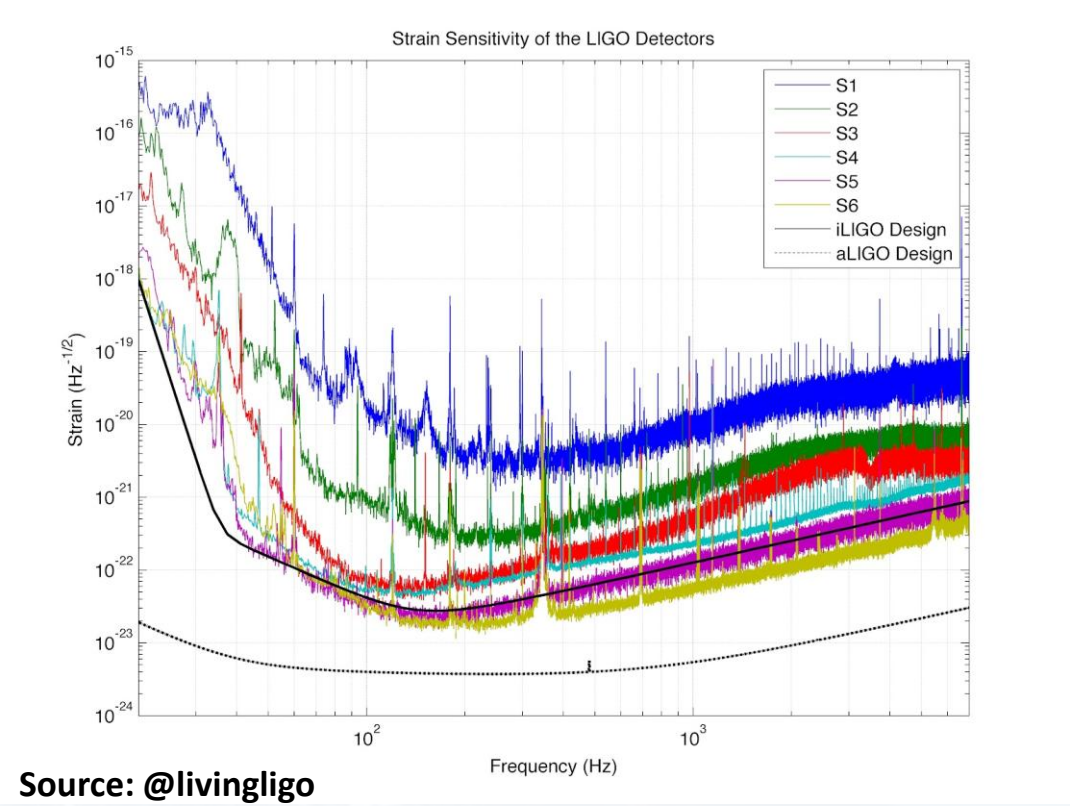


automating workflows (image to diagnosis, Chem 277B project, **Elle MacLennan, Carmen Matar, Timothy Nguyen, Brandon Ton**)

Why Numerical Methods?



Source: CERN/ATLAS

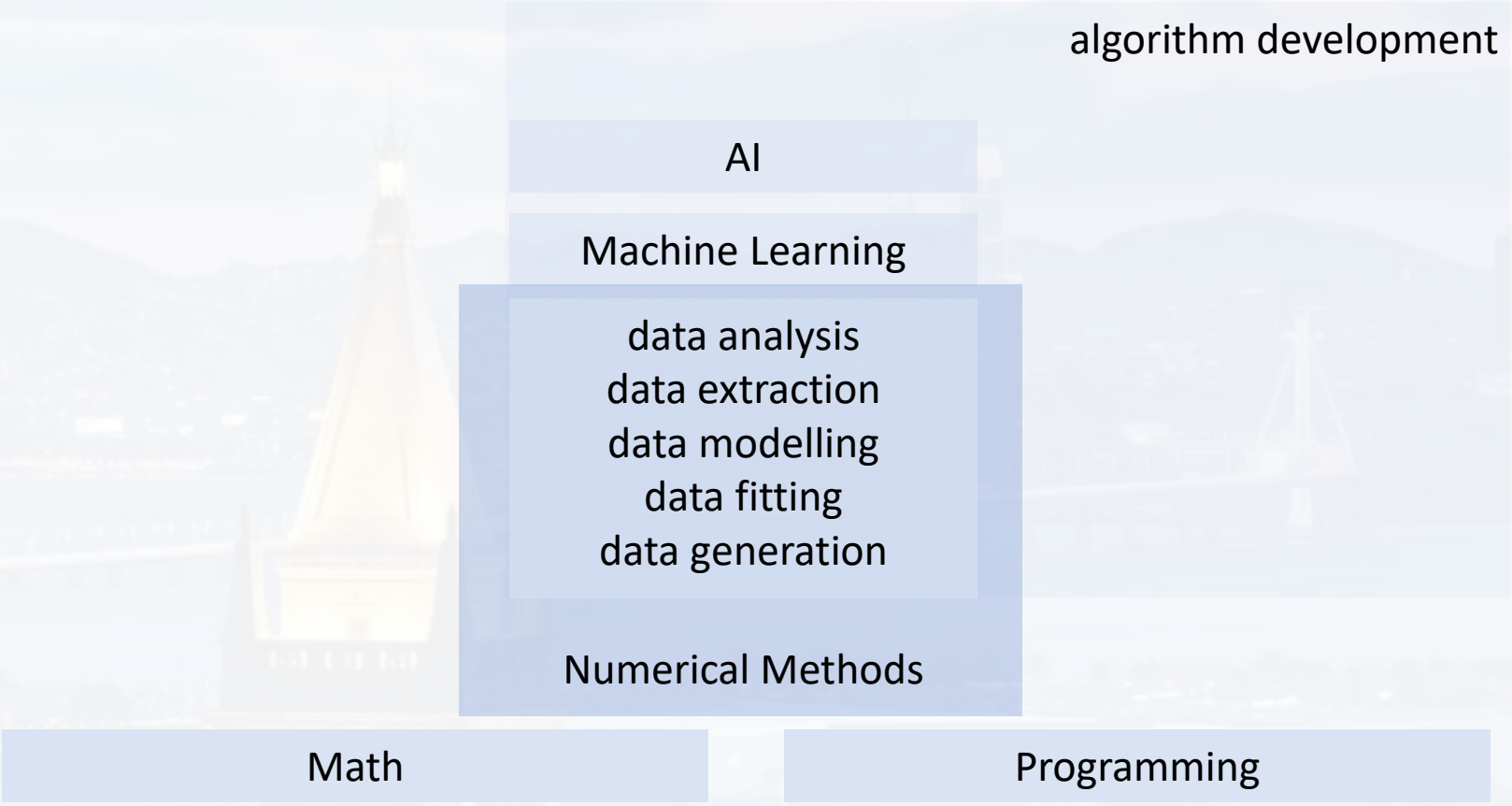


Source: @livingligo

- data analysis
- data extraction
- data modelling
- data fitting
- data generation

math + coding

Why Numerical Methods?



Why Computational Methods?

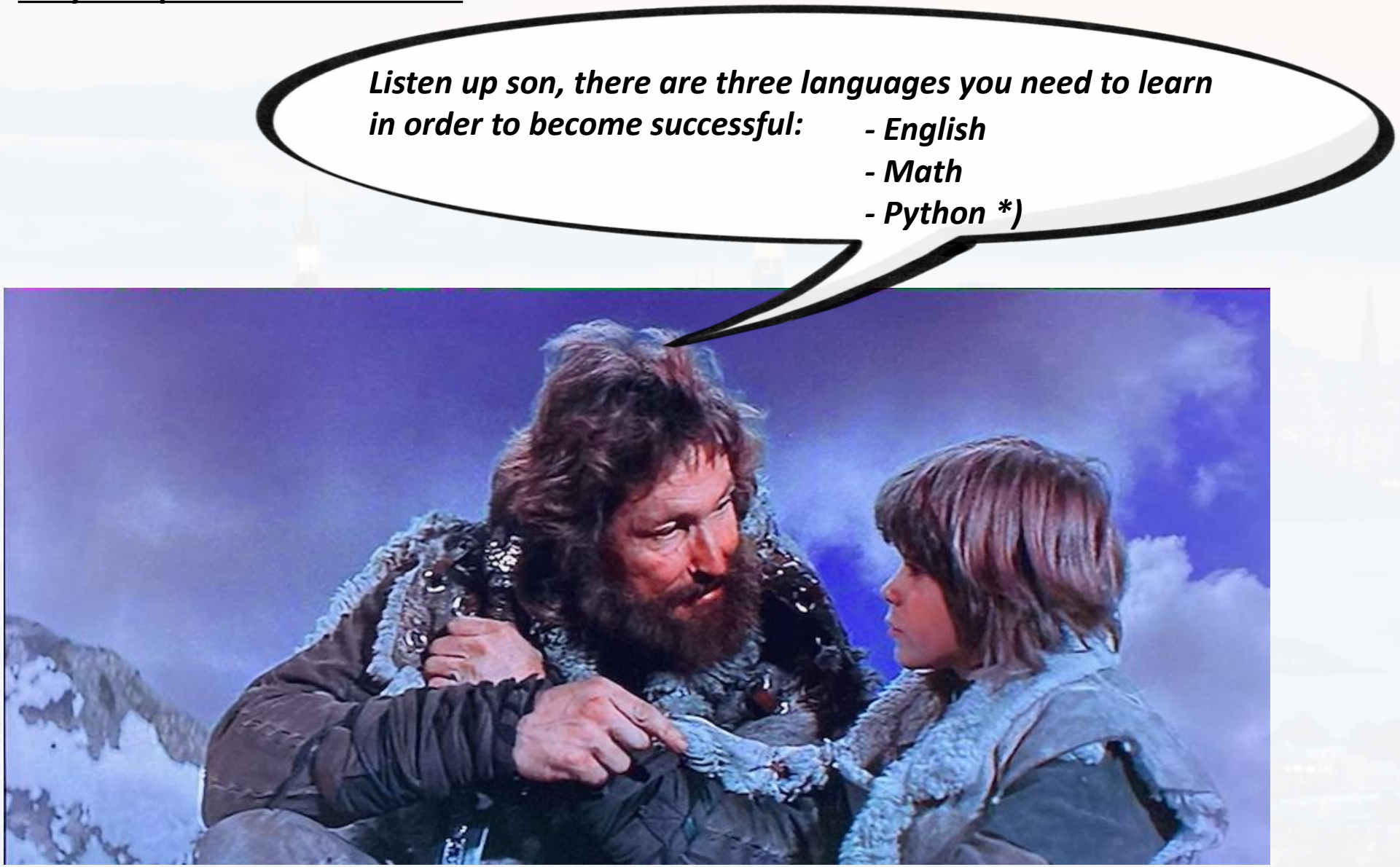
LinkedIn: 1st three Job Postings, Dec 19th 2025

Research Scientist - Vision Data Infrastructure
Storm3 • San Francisco, CA (Hybrid) • $250K/yr - $600K/yr • Vision benefit
Actively reviewing applicants
Promoted · Easy Apply

Staff AI Engineer
Lumicity • San Francisco Bay Area (Hybrid) • $250K/yr - $280K/yr • 4 benefits
Actively reviewing applicants
Promoted · Easy Apply

Staff AI/ML Engineer
Harnham • Sunnyvale, CA (On-site) • $200K/yr - $225K/yr • 4 benefits
Actively reviewing applicants
Promoted · Easy Apply

Why Computational Methods?



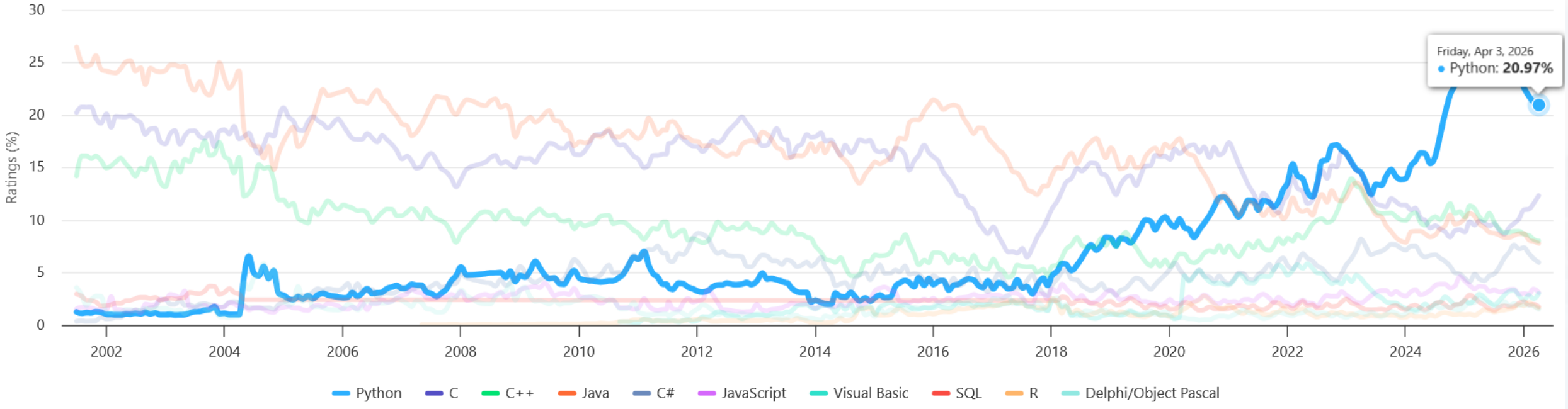
*) or C++

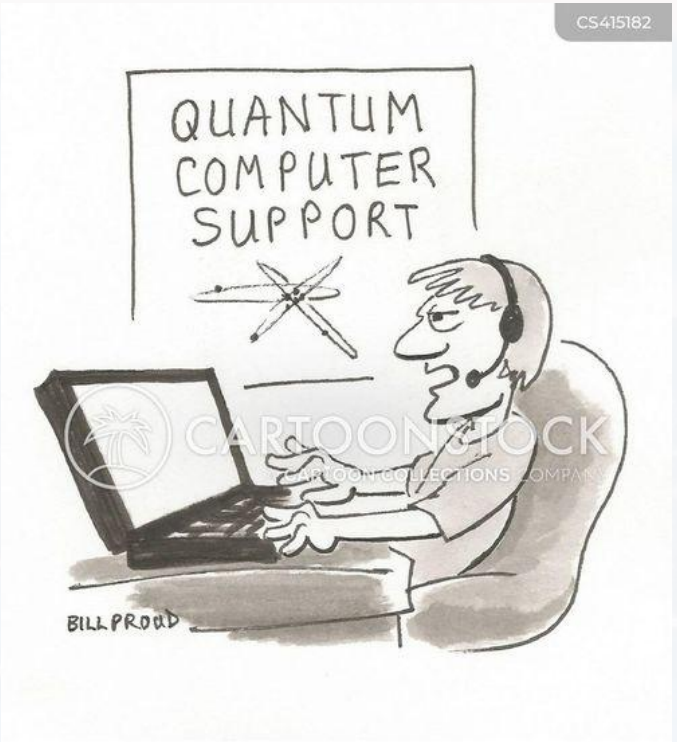
Introduction

TIOBE index, May 2026

Apr 2026	Apr 2025	Change	Programming Language	Ratings	Change
1	1		Python	20.97%	-2.11%
2	3	↑	C	12.34%	+2.39%
3	2	↓	C++	8.03%	-2.30%
4	4		Java	7.79%	-1.84%
5	5		C#	5.98%	+1.59%
6	6		JavaScript	3.11%	-0.60%
7	8	↑	Visual Basic	3.02%	+0.08%
8	10	↑	SQL	1.75%	-0.44%
9	14	⇈	R	1.62%	+0.43%
10	9	↓	Delphi/Object Pascal	1.52%	-1.01%
11	12	↑	Scratch	1.48%	+0.13%
12	19	⇈	Perl	1.48%	+0.57%
13	11	↓	Fortran	1.22%	-0.82%
14	13	↓	PHP	1.17%	-0.13%
15	7	⇊	Go	1.09%	-1.92%

TIOBE index, May 2026





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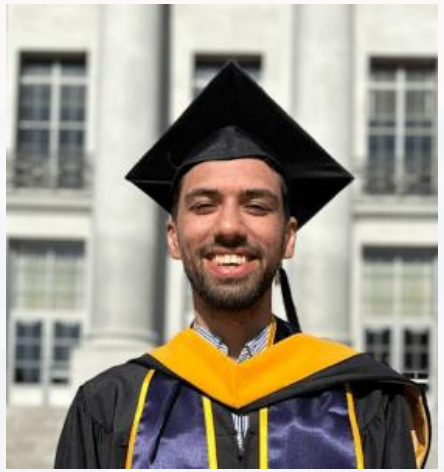
GSI: Usman Jamshed

Software Engineer at Schrödinger

MSSE Alumnus



Lecturer: Markus Hohle

Lecturer at UC Berkeley &

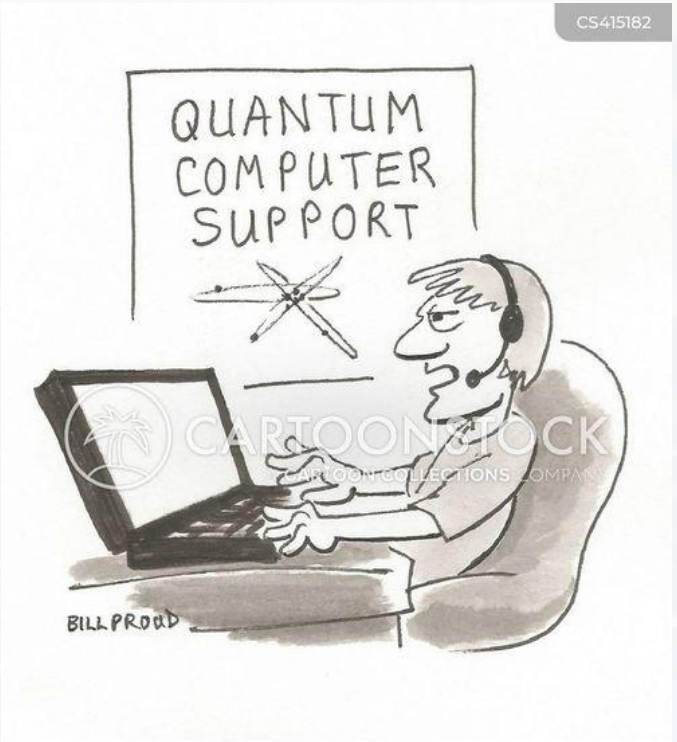
Data Analysis Consultant

PhD Physics

Introduction

Lecture (Markus):	Monday,	5:00 – 8:00pm PT
Discussion (Usman):	Wednesday,	5:30 – 6:30pm PT
Lab Session (Markus):	Wednesday,	6:30 – 7:30pm PT
Office Hours (Markus):	Friday,	5:00 – 7:00pm PT
Material:	Recorded Lectures:	bcourses
	codes/slides	bcourses & GitHub
	HW assignments	bcourses & GradeScope
Grades:	Problem Sets:	40%
	Programming Projects	20%
	Lecture Exercises	20%
	Discussion & Lab Participation	20%

Course Start Date:	**Monday May 18th, 2026**
Course End Date:	**Friday, August 14th, 2026**
Week 1:	Introduction to Scientific Computing and Python Libraries
Week 2:	Linear Algebra Fundamentals
Week 3:	Vector Calculus
Week 4:	Numerical Differentiation and Integration
Week 5:	Solving Nonlinear Equations
Week 6:	Probability Theory Basics
Week 7:	Random Variables and Distributions
Week 8:	Statistics for Data Science
Week 9:	Eigenvalues and Eigenvectors
Week 10:	Simulation and Monte Carlo Method
Week 11:	Data Fitting and Regression
Week 12:	Optimization Techniques
Week 13:	Machine Learning Fundamentals



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Introduction

1) reading files (.xlsx, .xls, .csv, .txt, …) pandas (standard)

+ analyzing/evaluating/manipulating data frames

faster than pandas, but fewer functions

- `dask`
- `polars`
- `fireducks`

Introduction

1) **reading files (.xlsx, .xls, .csv, .txt, …)** pandas (standard)

+ analyzing/evaluating/manipulating data frames

pandas	excel	104.00sec
polars	excel	10.17sec
pandas	csv	1.39sec
polars	csv	0.14sec
dask	csv	0.02sec

factor of 4000 – 5000!!

pandas

python

dask

polars

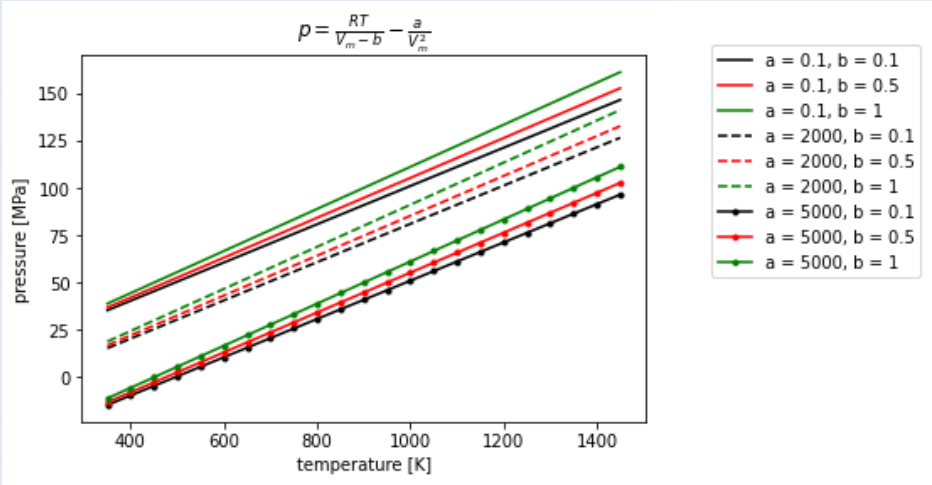
FireDucks

Introduction

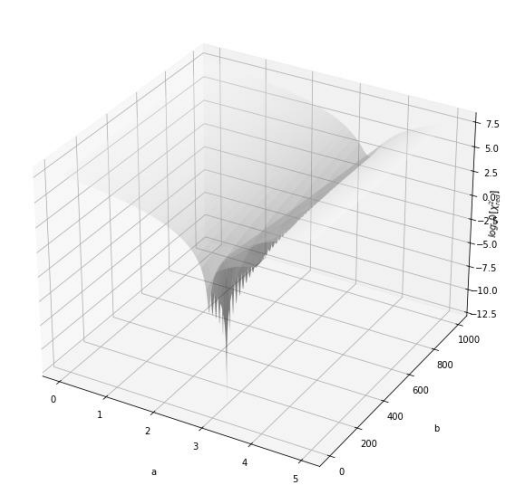
1) **reading files (.xlsx, .xls, .csv, .txt, …)** pandas (standard)

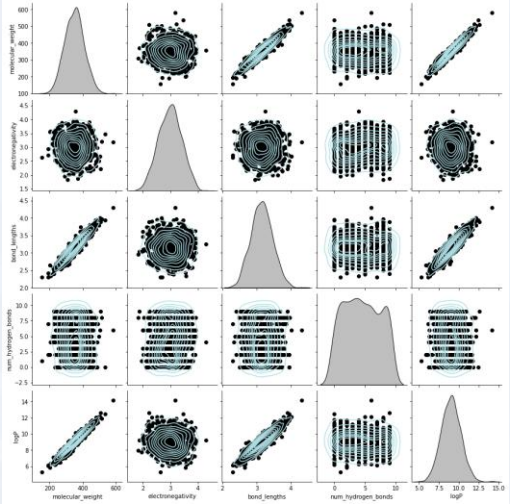
2) **plotting**

matplotlib



seaborn





Introduction

1) **reading files (.xlsx, .xls, .csv, .txt, …)** pandas (standard)

2) **plotting** matplotlib, seaborn

3) **numerical methods**

math

basics

```
math.
acos
acosh
asin
asinh
atan
atan2
atanh
```

numpy

standard

```
np.random.
hypergeometric
laplace
logistic
lognormal
logseries
mtrand
multinomial
```

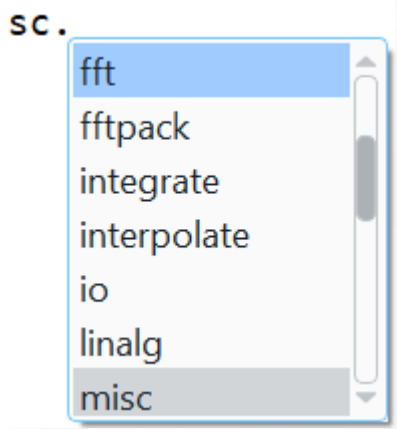
Introduction

1) **reading files (.xlsx, .xls, .csv, .txt, …)** — `pandas` (standard)

2) **plotting** — `matplotlib, seaborn`

3) **numerical methods**

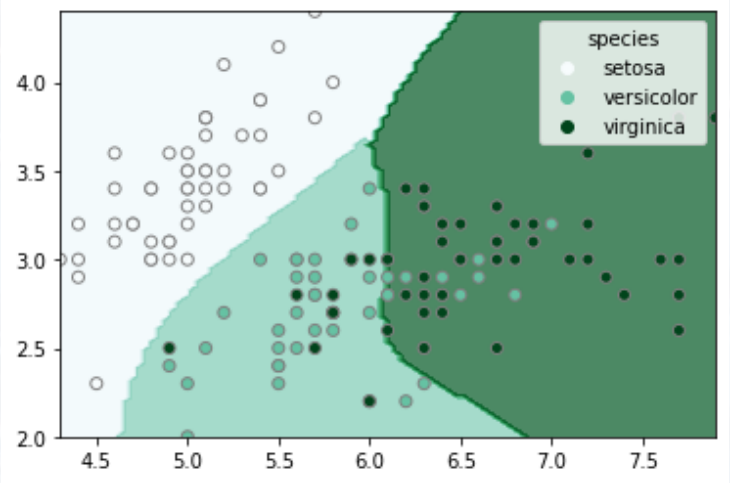
`math`		basics
`numpy`	NumPy	standard
`scipy`	SciPy	

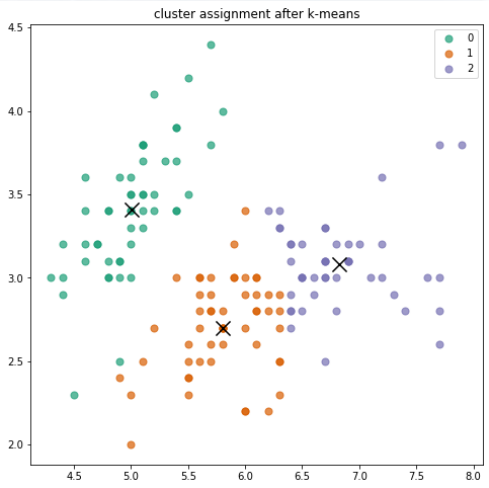


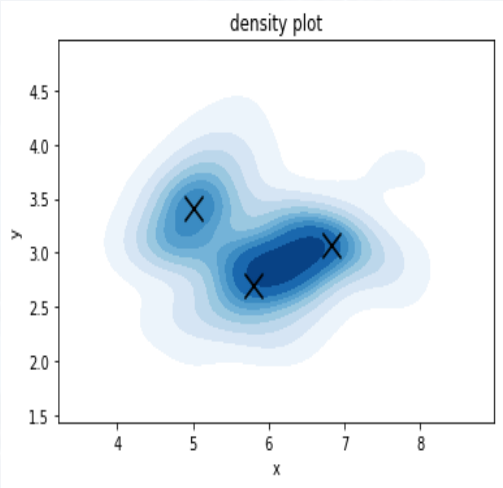
- num. integration/differentiation
- Fourier transformation
- optimization
- curve fitting …

Introduction

1) **reading files (.xlsx, .xls, .csv, .txt, …)** — `pandas` (standard)
2) **plotting** — `matplotlib, seaborn`
3) **numerical methods** — `math, numpy, scipy`
4) **machine learning** — `scikitlearn`





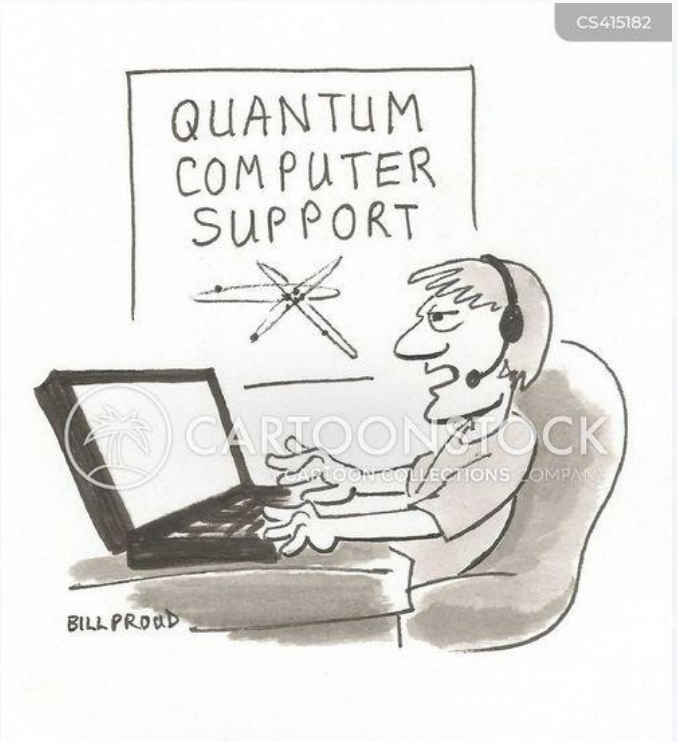


Introduction

Python Libraries

1) **reading files (.xlsx, .xls, .csv, .txt, …)**	`pandas` (standard)
2) **plotting**	`matplotlib, seaborn`
3) **numerical methods**	`math, numpy, scipy`
4) **machine learning**	`scikitlearn`
5) **ANN/AI/DeepLearning**	

TensorFlow Keras PyTorch



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- reading different file formats (same content)
- benchmarking
- coding warm-up for course
- FYI dynamic programming

see: `01_Lecture_Exercise.ipynb`

Introduction

- reading different file formats (same content)
- benchmarking
- coding warm-up for course
- **FYI dynamic programming**

see: `01_Lecture_Exercise.ipynb`

```
dfPandasCSV = ReadWithAnyToolAnyMethod()
dfPandasCSV = ReadWithAnyToolAnyMethod(filename = 'Data_set_0.xlsx', my_method = 'read_excel')

dfPandasCSV = ReadWithAnyToolAnyMethod(my_tool = 'dd')

dfPandasCSV = ReadWithAnyToolAnyMethod(my_tool = 'pl')
dfPandasCSV = ReadWithAnyToolAnyMethod(filename = 'Data_set_0.xlsx', my_tool = 'pl', my_method = 'read_excel')
```

```
Total runtime: 2.937999999994645 seconds
Total runtime: 198.5779999999795 seconds
Total runtime: 0.0 seconds
Total runtime: 0.1870000000053551 seconds
Total runtime: 17.0 seconds
```

Thank you very much for your attention!